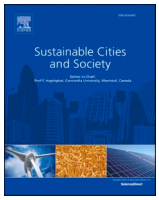
Sustainable Cities and Society 108 (2024) 105508

Contents lists available at ScienceDirect

Sustainable Cities and Society

journal homepage: www.elsevier.com/locate/scs



Urban flood susceptibility mapping using remote sensing, social sensing and an ensemble machine learning model

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ARTICLE INFO

Keywords:
Urban flood
Ensemble model
Machine learning
Remote sensing
Social sensing

ABSTRACT

Flood susceptibility mapping is crucial for urban disaster management. However, the heterogeneity of urban land use and the complexity of terrain pose challenges to the accuracy and generalizability of flood models. In this paper, we propose a framework named the EMF model for urban flood mapping. Specifically, the social sensing and remote sensing were used for flooding information collection. The XGBoost, Support Vector Classifier (SVC), Multilayer Perceptron (MLP), and Multimodal Deep Learning (MDL), were used as the predictive models, and their results were ensembled using a Random Forest model to produce the final outcome. Results show that the EMF model outperforms the standalone models, with an accuracy of 0.942 on the training set and 0.940 on the testing set. The accuracy of the five models is ranked as EMF > MDL > XGBoost > SVC > MLP. The flood map indicates that flooding has a greater impact on the outskirts and suburbs of the city compared to its central urban areas. Farmland is the most affected land type, making up 54.8 % of the flooded area. Overall, the proposed framework enables rapid and accurate identification of flood-prone areas, providing technical support for managers in formulating effective flood prevention strategies.

1. Introduction

In recent years, frequent occurrences of extreme precipitation events have led to severe urban flooding disasters, resulting in property damage, infrastructure interruptions, and posing threats to public safety (Tellman et al., 2021). For instance, on July 21, 2012, Beijing experienced a catastrophic flood event that resulted in more than 70 deaths and impacted 1.6 million people (Wang et al., 2020). A reliable urban flooding monitoring method is of utmost importance for the government to formulate effective risk management and disaster mitigation strategies. However, accurately mapping the spatial inundation areas of flooding in urban areas poses challenges. Firstly, traditional flood monitoring relies heavily on in-situ observations, which are unable to supply real-time comprehensive data across extensive regions and may also expose on-site researchers to potential risks (Chen et al., 2019; Merkuryeva et al., 2015). Secondly, the performance of flood models exhibits significant variability when applied in urban areas due to the heterogeneity of urban land cover and the complexity of terrain (Padulano et al., 2021).

Hydrological and hydraulic models are widely used tools for assessing urban flood risk (Kim et al., 2023; Piyumi et al., 2021). These physics-based methods require extensive high-quality data, such as rainfall data, basin characteristics, and soil types, and the model parameters are complex, limiting their generalizability. Due to the high-resolution and all-weather imaging capabilities, the Synthetic Aperture Radar (SAR) satellite is getting more and more attention in the flood monitoring (Chew et al., 2023). The lower backscatter properties of flooded regions in SAR images can be utilized to distinguish flooded areas from other ground objects. By utilizing the characteristic, a variety of data-driven based methods have been developed to map urban flood susceptibility regions (Amitrano et al., 2018; Martinis et al., 2015). The data-driven based methods can be divided into empirical methods and machine learning (ML) methods. The commonly used empirical methods, such as histogram thresholding and texture analysis, identify flood areas through intensity thresholding and spatial pattern analysis. These methods have lower data requirements than the physical-based methods and have been applied in many studies with fairy accuracy (Hao et al., 2021; McCormack et al., 2022). However, uncertainties can still arise in their application for urban flooding extraction, primarily stemming from issues related to rain-mixed pixels, sensitivity to lighting and shadow effects (Luo et al., 2022).

Given the strong ability to discern intricate patterns and extract

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https://doi.org/10.1016/j.scs.2024.105508
Received 16 January 2024; Received in revised form 15 April 2024; Accepted 3 May 2024
Available online 5 May 2024
2210-6707/© 2024 Published by Elsevier Ltd.

features, ML methods can effectively alleviate the shortcomings of empirical models (Choubin et al., 2022; Guo et al., 2023). The commonly used ML model, including tree models, support vector models, and neural network models. These models have demonstrated potential in generating flood susceptibility maps in the previous studies (Costache et al., 2020; Hosseini et al., 2020; Taromideh et al., 2022; Xie et al., 2021). For instance, Rafiei-Sardooi et al. (2021) compared the performance of SVM, Random Forest classification (RFC), and Boosted Regression Tree (BRT) for urban flooding areas extraction based on socio-economic data. The results indicated that RFC achieved the highest accuracy, reaching 0.958 on the test dataset. Ma et al. (2021) proposed a flash flood risk assessment framework based on XGBoost model. The accuracy of the model for flooding area identification in the testing dataset was 0.84. Mehravar et al. (2023) developed a flood detection method based on SVR and Sentinel-1 images. The results indicated that the SVR model exhibited good performance for flood susceptibility mapping, with an accuracy of 0.81. Although the ML method has shown enormous potential in modeling flood hazard maps, these methods still exhibit certain limitations. A single model may not adequately learn the features of input data, resulting in limited accuracy in flood identification under complex urban land cover and terrain conditions (Pham et al., 2021; Rasheed et al., 2022). Additionally, the accuracy and effectiveness of the ML methods heavily hinge on the quantity and quality of data (Guo et al., 2020). Identifying accurate samples from flood-affected areas can be challenging and further constrains the performance and applicability of these models.

Social sensing has been demonstrated as an effective strategy for providing real-time disaster information (Rosser et al., 2017; Smith et al., 2015). It involves the collection of disaster-related text data from sources such as field data, social media, and the internet. By combining social sensing and remote sensing, a more comprehensive datasets related to flood locations can be obtained. For model selection, there is no universally optimal flood model because of the diverse and complex nature of flood influences across regions (Tehrany et al., 2014). The hybrid model has the capability to adapt to complex data structures and multimodal distributions (Zhu et al., 2023). By integrating the predictive capabilities of multiple models, hybrid model holds the promise to overcome the limitations of single models in accurately extracting urban flood extents and further improving their generalizability.

In summary, the scarcity of data and inadequate model generalization remain bottlenecks that restrict the performance of flooding models for accurately mapping urban flood inundation areas. To address the aforementioned issues, this study has the following objectives: (1) proposed a scheme of higher reliability for flooding area extraction based on social sensing data, Sentinel-1 GRID images and an ensembled ML model named the “Ensemble - Random Forest” (EMF) model. Specifically, four state-of-the-art ML models, namely XGBoost, SVC, MLP, and MDL, were independently employed for flood area extraction. Subsequently, the outputs of these four models were fused using the RFC model to identify the final flood-prone areas; (2) validate the performance of the EMF model by conducting a comparative analysis of its accuracy against that of four individual models. (3) Map the spatial patterns of the flood region during a severe rainstorm event that occurred in Tianjin, China. Identify the primary locations of urban flooding and investigate the relationship between their distribution and the underlying surface of the city. Overall, this study is expected to significantly enhance flood disaster monitoring capabilities, providing technical support for governmental flood emergency management and the realization of sustainable urban development.

2. Study area and data description

2.1. Study area

Tianjin is a bustling coastal metropolis with a population of over 15 million, sits within the fertile North China Plain. Covering an area of 11,966.45 km^2, the city forms the estuary of the vital Yong Ding River, a key tributary of the Hai He River Basin (Chen et al., 2023). Previous studies indicated that Tianjin is the most vulnerable city with the highest risk of flood exposure (Li et al., 2022; Wu et al., 2021). During the intensive rainstorm period, the continuous flow of river water originating from Beijing and Northern China, which passes through Tianjin before reaching the Bohai Sea causing the risk of riverine flooding to the city. Moreover, the increased impervious surfaces due to the rapid urbanization hinders the infiltration of surface runoff, thereby elevating the risk of surface water flooding. Historically, Tianjin experienced extraordinary flood events in 1963, 1996 and 2012 (Kron et al., 2012; Zhao et al., 2018). More recently, an intensive rainstorm occurred on July 31, 2023, resulting in widespread flooding in various areas of Tianjin and causing disruptions in urban transportation and significant property damage (Zhang et al., 2023). Accurate estimations of flood extent are crucial for local governments to effectively provide public health aids, material support, and infrastructure repairs. Therefore, this study selected Tianjin as a representative area to evaluate the proposed method’s effectiveness in mapping urban flooding (Fig. 1).

2.2. Satellite data

In this study, Sentinel-1 Synthetic Aperture Radar (SAR) data was used to extract the urban flooding area. Sentinel-1 is a publicly accessible radar dataset with a high resolution of 10 m, allowing for large-scale hydrological analysis. The calibrated and ortho-corrected Sentienl-1 SAR data corresponding to the time of the flooding event on July 31 2023 was obtained directly from the Google Earth Engine (GEE) platform. The image collection of the Sentinel-1 SAR data in GEE platform has been filtered by using Interferometric Wide (IW) swath mode, orbiting pass, ascending, and descending options. A speckle filter was subsequently applied to the imagery to suppress noise pixels. Two single channels named like-polarized (VV) and cross-polarized (VH) were extracted from the Sentinel-1 SAR data, The quantity of backscatter signal detected by the sensor from ground features is influenced by the structure of the object or feature. The amount of backscatter observed by the sensor may differ depending on the polarization. Previous studies have shown that the combination of different polarization may provide more effective information for urban flooding extraction (Henry et al., 2006). Thus, apart from the two single channels (VV and VH), ten polarization combinations were derived from VV and VH were used in the study to enhanced the flooding area information (Table 1).

In clear-sky conditions, optical satellite imagery offers a distinct advantage in land cover classification, enabling precise discrimination of surface features through spectral differentiation (Kutser et al., 2020). In this study, the Sentinel-2 imagery was employed for permanent water body extraction. The Sentinel-2 imagery contained 13 spectral channels with the maximum spatial resolution of 10m. The cloud-free Level-2 products of Sentinel-2 from June 1 to July 30, 2023 were used obtained from GEE platform. All the Level-2 products had been processed by the sen2cor atmospheric correction algorithm. The two SWIR channels were resampling to 10m. The dynamic surface water extent (DSWE) model was employed to obtained the surface water extents. The DSWE algorithm calculates five water indices using the visible bands, NIR, and SWIR, NIR channels, and then delineates water bodies based on varying thresholds applied to different water indices. For the detail of the DSWE algorithm, please reference Petrakis et al. (2022).

2.3. Flood influencing factors

Land use and topographical factors play crucial roles in shaping the occurrence of urban floods. For example, Normalized Difference Vegetation Index (NDVI) reflects vegetation cover, which influences infiltration and surface runoff, impacting flood potential. Altitude amplifies flood risks through steeper slopes, higher precipitation, altered temperatures, and unique vegetation patterns, impacting both settlements

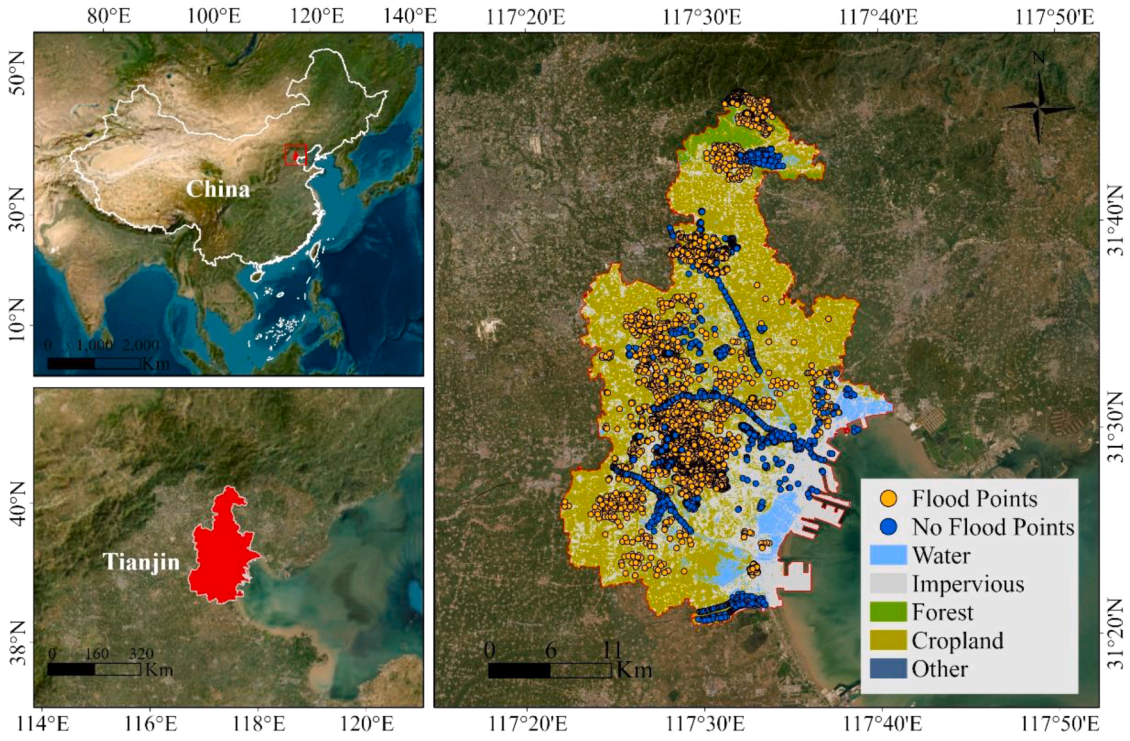

Fig. 1. Location of the study area. (a) – (b) The geographic location of Tianjin in North China. (c) Tianjin's land use map in 2021 and the distribution of sampling points. Blue dots represent flood points, orange dots represent non-flood points. The land cover data was obtained from the https://zenodo.org/record/5816591.

Table 1
Various polarization combinations in Sentinel-1 SAR data.

Sequence number	Feature name	Sequence number	Feature name
1	VV	7	VV * VH
2	VH	8	VV / VH
3	log|*VV*|	9	$VV^2 + VH^2$
4	log|*VH*|	10	$VV^2 - VH^2$
5	VV + VH	11	$VV^2 * VH^2$
6	VV-VH	12	VV^2 / VH^2

and downstream areas (Chaulagain et al., 2023). Distance to stream is associated with flood risk due to rapid runoff accumulation and direct exposure to overflowing waterways. Steeper slopes may accelerate water flow, intensifying runoff and flood severity (Sunada & Hong, 1988). Based on the previous study and data availability (Li et al., 2023), two land use factors, including NDVI and land cover, and five topographical factors, including distance to stream, slope, altitude, topographic wetness index (TWI), and aspect, were initially selected in this study (Fig. 2). The topographical datasets used in this study were produced from the 30 m SRTM DEM imagery. The NDVI map was generated from Landsat8-OLI imagery acquired in the year 2023 using the following equation.

$$NDVI = \frac{Band_{nir\ red} - Band_{red}}{Band_{nir\ red} + Band_{red}} \quad (1)$$

Where $Band_{nir\ red}$ is the near-infrared band, and $Band_{red}$ is the red band in Landsat8-OLI imagery.

Building on the flood influencing factors and Sentinel-1 SAR-derived polarization combinations, we employed the Pearson correlation analysis to explore their relationship with flooding. Subsequently, the random forest feature importance ranking (FFIR) method was employed to select input variables from the most significant features for model development. The FFIR method is commonly used for data preprocessing to reduce overfitting by quantifying the influence of each variable on the model's decision-making process.

3. Methods

3.1. Social sensing

Due to the richness of information, near real-time transmission channels, and low-cost data generation, social media platforms are garnering increasing attention in the field of disaster emergency management. In this study, a social sensing framework proposed by Lin et al. (2023), was employ to gather relevant textual information related to flooding events from social media posts. Four social media platforms, namely, Toutiao, YouTube, TikTok, and Kuaishou, were selected as the data sources. Toutiao serves as an official platform for authoritative institutions to disseminate information, while YouTube, TikTok, and Kuaishou primarily function as platforms for user-generated content and self-media publishing. A keyword system related to flooding events, including location names (Tianjin), flood-related terms (e.g., floodwater, inundation, submerged), and event-specific details (e.g., date and affected areas) was established to retrieve social sensing data on these social media platforms. The Keywords were updated manually based on their frequency of appearance in social media posts. The type, coverage, brief introduction, and web link of the social media posts were subsequently automatically retrieved via the Beautifulsoup4 (version 4.11.1) and Selenium (version 4.8.0) Python libraries. Once the social media posts related to floods were identified, time and location information within the posts were acquired through visual interpretation, and subsequently integrated with digital maps to obtain precise geographical coordinates.

An example of the social sensing data acquisition process is depicted in Fig. 3. To ensure the reliability of the acquired social sensing data, we filter out false data based on the following three criteria: (1) The heavy rainfall in Tianjin occurred from July 28, 2023, to August 1, 2023. Given the lag time associated with flooding, the timestamps of social media posts must precede August 3, 2023. (2) Flood scenes should be clearly

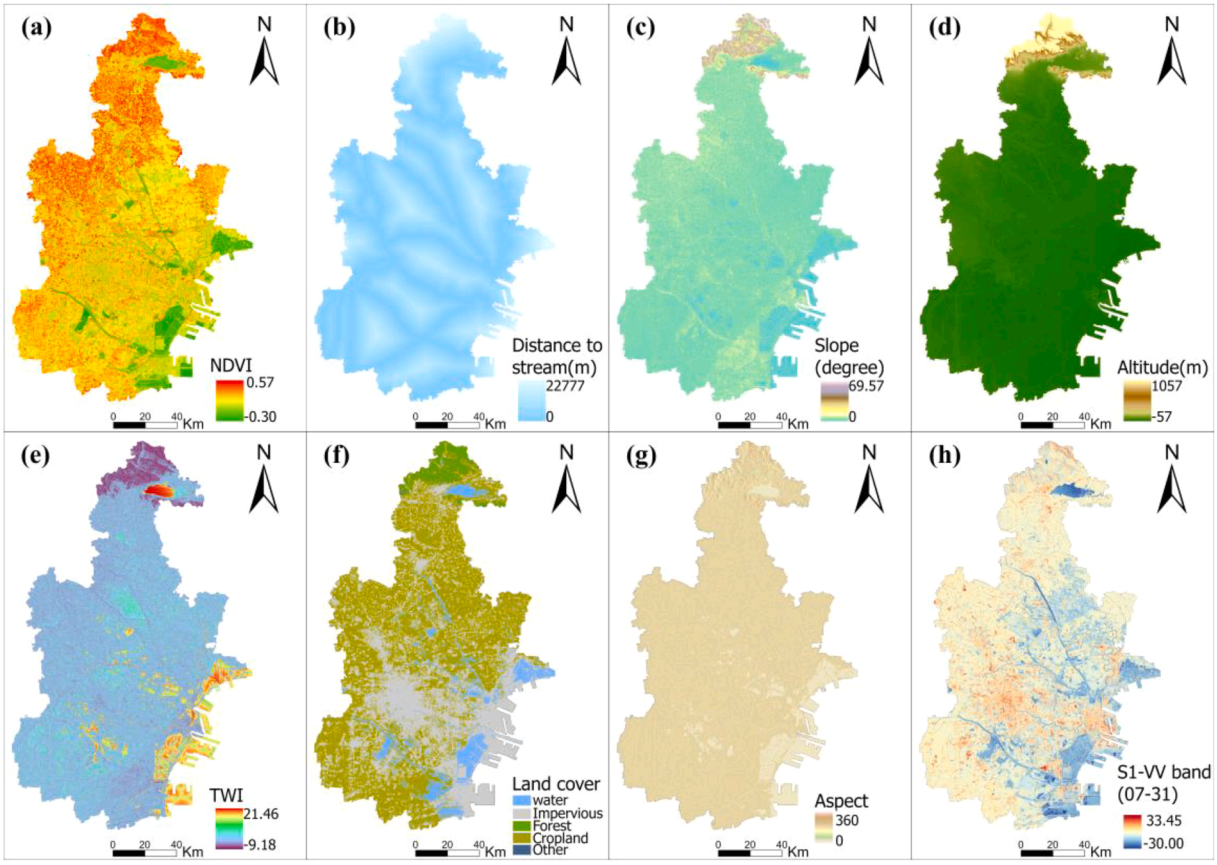

Fig. 2. Land use and topographical datasets used in this study. (a) NDVI, (b) Distance to stream, (c) slope, (d) altitude, (e) TWI, (f) land cover, (g) aspect, (h) Sentinel-1 VV polarization band.

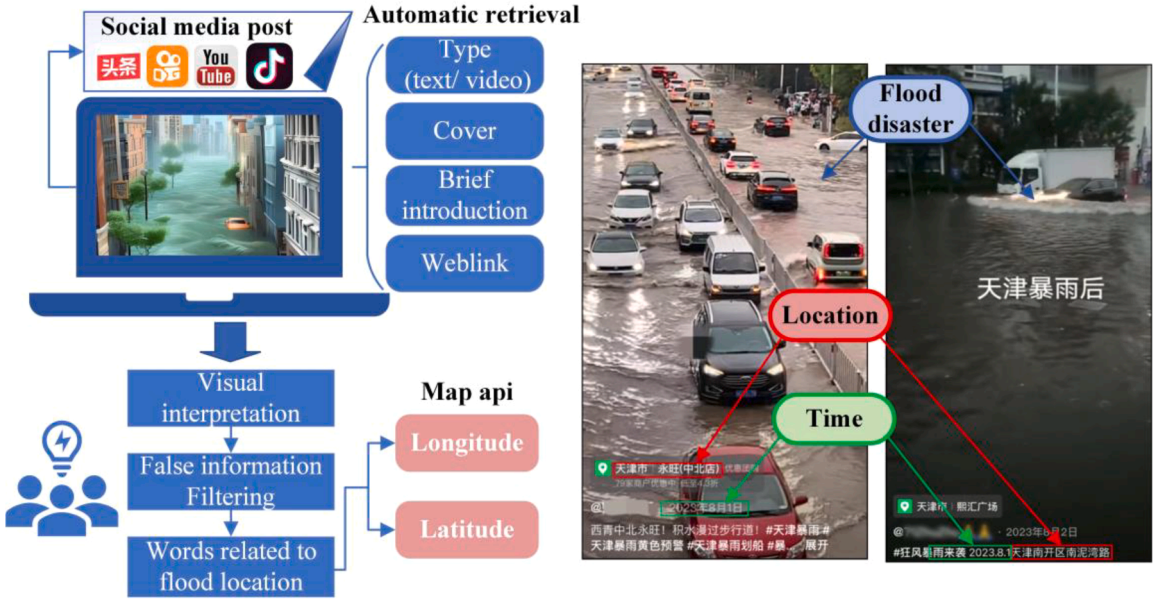

Fig. 3. An example of the social sensing data acquisition process.

observable in the social media posts. (3) The posts from Toutiao must originate from official media sources such as the Tianjin Meteorological Observatory and Tianjin Traffic Police. The personal account homepages of self-media users on platforms such as TikTok and Kuaishou must include posts related to daily life in addition to those concerning floods.

3.2. ML models

In this study, an ensemble model called EMF was developed for Urban flood susceptibility mapping. The EMF model consists of four sub-models and one ensemble model. The sub-models include XGBoost, SVC,

MLP, and MDL, chosen as representatives of tree-based models, support vector models, and neural networks. Furthermore, the RFC was chosen as the ensemble model to integrate the predictions of the sub-models to generate the final model output.

XGBoost is an efficient gradient boosting tree algorithm, suitable for processing structured data and classification problems. The important hypermeters in XGBoost include n_estimators, subsample, learning_rate, reg_lambda, reg_alpha and max_depth. The n_estimators and max_depth parameters determine the number and maximum depth of trees in the model, respectively, influencing its complexity and stability. The Subsample parameter specifies the proportion of samples used in each iteration, aiding in controlling overfitting. The learning_rate parameter controls the step size of parameter updates during each iteration, affecting convergence speed and generalization capability. The reg_lambda and reg_alpha parameters regulate the strengths of L2 and L1 regularization, respectively. XGBoost was executed via Python 3.7, employing the xgboot library (version 1.2.1). The parameters for XGBoost were set as follows: n_estimators to 100, subsample to 0.8, learning_rate to 0.01, reg_lambda to 1, reg_alpha to 0, and max_depth to 3. The objective function of XGBoost is as follows:

$$Objective_{XGB} = \sum_{i=1}^{N} l(y_i, \hat{y}_i) + \Omega(f) \tag{2}$$

Where N is the number of samples, y_i is the true label of the ith sample, $\hat{y}_i$ is the model's predicted output, $l(y_i, \hat{y}_i)$ is the loss function and $\Omega(f)$ is a regularization term.

SVC excels in handling complex data and high-dimensional feature spaces, possessing strong generalization capabilities. The hyperparameters considered for fine-tuning SVC encompassed kernel, penalty parameters (C), gamma, degree, and cof0. The Kernel parameter specifies the type of kernel function used to transform data into a higher-dimensional space, impacting the model's capacity to handle feature relationships. The C parameter was used to balances the penalty applied to misclassifications. The gamma parameter was used to controls the influence range of individual data points within the kernel function. The degree parameter determines the degree of the polynomial mapping, influencing model complexity. The Coef0 parameter represents the constant term in kernel functions, influencing the bias of the decision function. The utilization of SVC was executed via Python 3.7, employing the Scikit-learn library (version 1.0.1). The parameters for SVC were set as follows: kernel to "rbf", C to 1, gamma to "scale", degree to 3, and cof0 to 0. The objective function of SVC is as follows:

$$Objective_{svc} = min_{\omega,b}\frac{1}{2}\| \omega \|^2 + C\sum_{i=1}^{N}\max(0, 1 - y_i \cdot f(x_i)) \tag{3}$$

Where ω and b are the model parameters, $\| \omega \|^2$ is the squared norm of the weights, C is the regularization parameter that controls the strength of regularization, y_i is the true label for the sample x_i, $\cdot$ denotes the dot product of vectors, and $f(x_i)$ is the model's output for the sample x_i.

By introducing hidden layers and activation functions, MLP is capable of learning and modeling complex structures and nonlinear relationships within data. MDL can effectively integrate multiple data types and information sources, enhancing model performance and generalization capabilities. The utilization of MLP and MDL models were implemented by TensorFlow (version 2.13.0). The stochastic gradient descent (SGD) was chosen to update models' parameters. An activation function named ReLU was added to each hidden layer to implement a nonlinear transformation of data. The sigmoid activation function was employed in the output layer to map output values to probability values. The initial learning rate of the MLP and MDL was set to 1e-3 and the loop for training was 1000 epochs. The objective function of MLP and MDL are as follows:

$$Objective_{Model} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{C=1}^{C} y_{i,c}\log\left(Model(x_i)_c\right) \tag{4}$$

RFC can effectively reduce the risk of overfitting in individual models and enhance the model's generalization capability by aggregating the predictions from multiple sub-models through voting or averaging. RFC was implemented by using the Scikit-learn library (version 1.0.1). The hyperparameters of RFC includ n-estimators, max depth and max features. The n-estimators and max depth parameters have the same meaning in XGBoost. The max feature parameter is utilized to enhance model diversity and mitigate overfitting by controlling the number of features considered for splitting at each node. The parameters for RFC were set as follows: n_estimators to 100, max depth to 3, max features. to "auto". The objective function of the EMF model is as follows:

$$Objective_{EMF} = \sum_{i=1}^{N}(L_{RFc}(XGB(x_i),\ SVC(x_i),\ MLP(x_i), MDL(x_i)),\ y_i) \tag{5}$$

Where N is the number of samples, L_{RF} is the loss function of the RFC model, $XGB(x_i),\ SVC(x_i),\ MLP(x_i), and\ MDL(x_i)$ are the predictions of the four submodels on the sample x_i.

3.3. Model development

The technological flowchart of this study was shown in Fig. 4. It consists of three steps:

(1) Dataset preparation. The geographical coordinates of the flooded areas were acquired using the social sensing method. Simultaneously, we utilize the ECMWF Reanalysis 5 (ERA5) datasets to identify dates with no recorded precipitation during the months of June and July in the Tianjin. The GEE and visual interpretation were used to identify samples of non-flooded areas, and the social sensing was used to gather the geographical coordinates of flood locations. Next, the GEE map API was used along with the collected geographical coordinates to retrieve the polarization information of Sentinel-1 SAR data for flood samples ($N = 1500$), water samples ($N = 1500$), and non-flooding samples ($N = 2000$). The spatial resolution of the Sentinel-1 SAR data was resampled to 30 m × 30 m in order to keep consistent with that of land use and topographical data. The mean values of pixels within 3 × 3 windows centered around sample points were extracted from down-scaled Sentinel-1 SAR satellite data and land use and topographical data. To eliminate redundant features, we conducted a correlation analysis between input features and floods to identify those significantly associated with flooding. To further reduce the complexity of the ML models and improve their accuracy, the FFIR method was used for identifying the optimal input features from those significantly correlated with floods. Finally, the dataset with the optimal features was divided into a training set and a testing set for model development, with a split ratio of 70 % and 30 %, respectively.

(2) Ensemble model construction. An ensemble ML model named EMF was proposed by using the generated dataset, the architecture of the EMF Model as shown in Fig. 5. Specifically, the training set was used to training four individual model named XGBoost, SVC, MDL and MLP. The RFC model was subsequently used to fit the residuals between the predictions from four models and the real values. The testing set was used to further verify the generalization ability of the model. For comparison, the result was benchmarked with the standalone machine learning models (XGBoost, SVR, MDL, and MLP).

(3) Urban flood extent extraction. The flooding susceptibility region was mapped by applying the trained EMF model to the Sentienl-1 SAR image on 31st July. After classifying the flooding

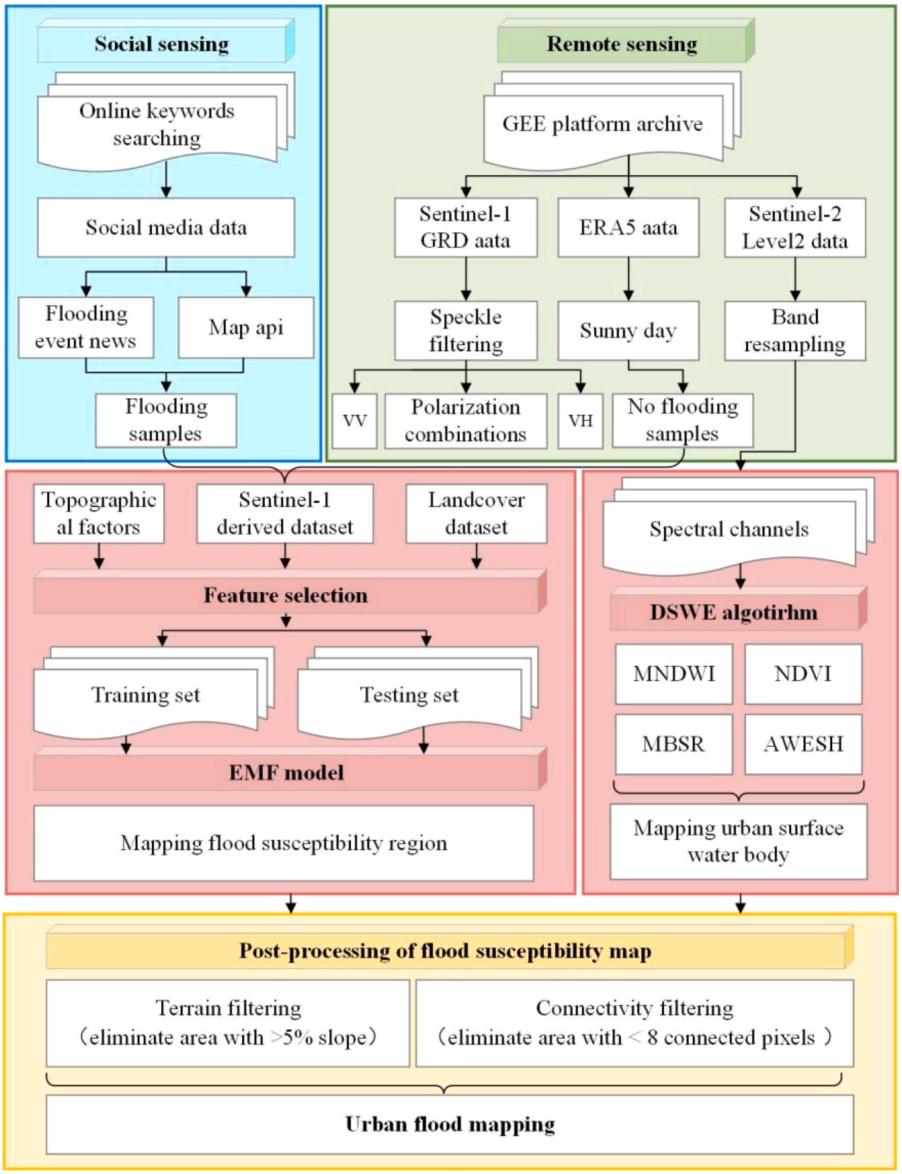

Fig. 4. Technical framework for urban flood area extraction based on remote sensing, social sensing, and EMF model.

susceptibility region using the EMF model, we applied the DSEW algorithm to the Sentinel-2 image to obtain permanent water areas. The preliminary extent of urban flooding was determined by subtracting the permanent urban water areas from the obtained flooding susceptibility region. All areas with a slope greater than 5 % in the preliminary urban flooding map were excluded using the terrain filtering method (Tazmul et al., 2022), as pixels with slopes exceeding this value will not retain water and will instead drain to lower elevation areas. The connectivity filtering method was used to exclude extremely small flooding region. The final urban flood extent was extraction after applying these two refinements.

3.4. Performance evaluation metrics

Five common evaluating metrics including accuracy, recall, precision, miss rate (MR) and F1 score were used to evaluate the performance of the developed ensemble machine learning model. Accuracy provides an overall measure of correct classifications, while recall emphasizes the importance of identifying all relevant instances, and precision focuses on the precision of positive predictions (Yang et al., 2019; Zhang et al., 2019). MR measures the ratio of incorrectly classified positive instances to the total number of actual positive instances (Shastry et al., 2023). It provides insight into the model's capability to detect all positive instances accurately. F1 score represents the harmonic mean of precision and recall, offering a holistic assessment of the model's precision and robustness (Tepe et al., 2023). The metrics expressions are shown in the following equations.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{5a}$$

$$Recall = \frac{TP}{TP + FN} \tag{6}$$

$$Precision = \frac{TP}{TP + FP} \tag{7}$$

$$MR = \frac{FN}{TP + FN} \tag{8}$$

$$F1\ score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{9}$$

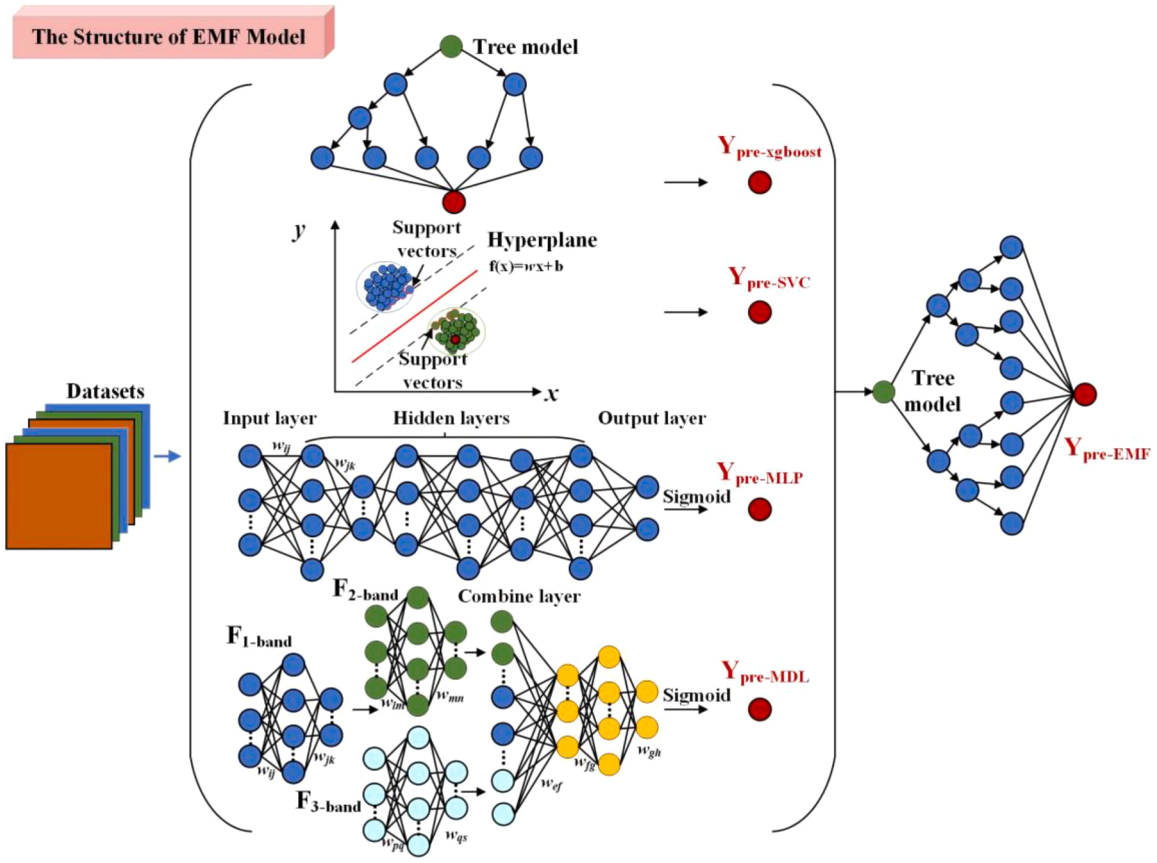

Fig. 5. The structure of the EMF Model

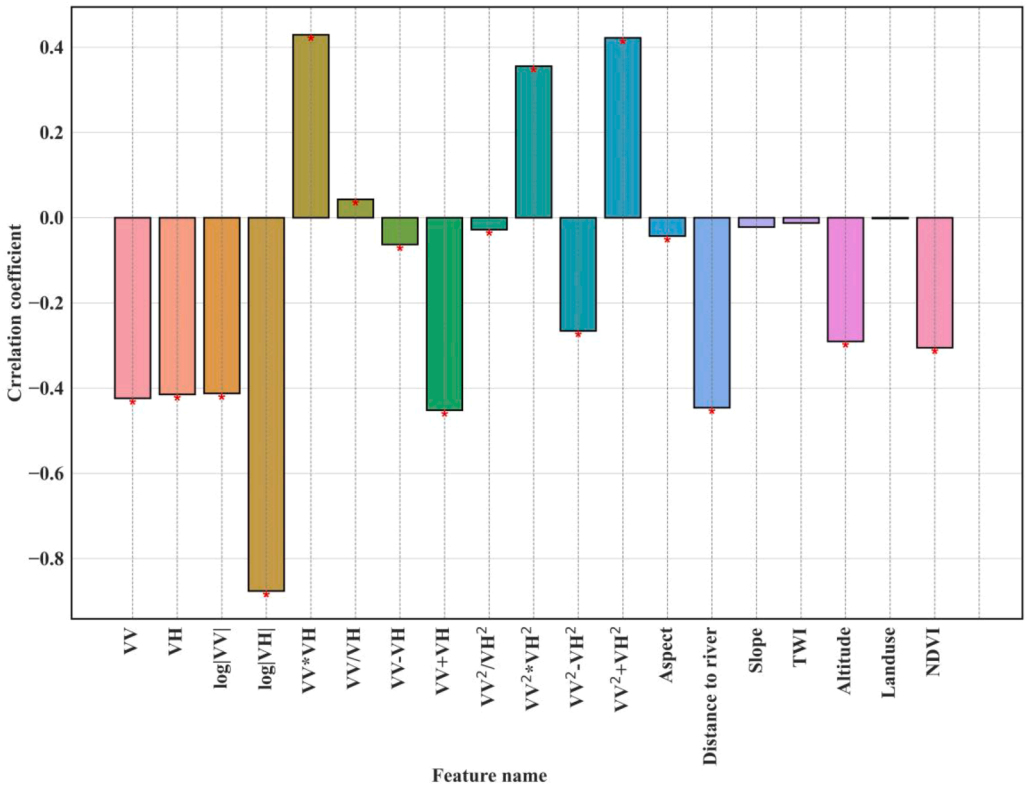

Fig. 6. The correlation of each input feature with the occurrence of flooding, with asterisks denoting significant correlation ($p < 0.05$).

where TP (true positives) is the number of pixels that the model correctly classifies as flood regions; TN (true negatives) represents the number of pixels that the model correctly classifies as non-flood areas; FP (false positives) represents the number of non-flood pixels wrongly classified as flood regions; FN (false negatives) represents the number of flood pixels wrongly classified as non-flood regions.

4. Results

4.1. Model feature selection

Exploring the correlation between input features and flooding occurrence is critical for understanding the flooding dynamics. It can also guide the selection of the most informative features and eliminate redundant or irrelevant ones. The correlation between input features and flooding as shown in Fig. 6. According to Fig. 6, the polarizations band of VV*VH, VV^2*VH^2, and VV^2+VH^2 demonstrate positive associations, while all other features exhibit negative relationships with flood occurrence. In general, the negative correlation between Sentinel-1's polarization bands and flood occurrence can be attributed to the strong absorption of polarized microwaves by water bodies. As a result, areas covered by water exhibit reduced VV and VH polarization signals. Distance to river shows the negative relationship with flood occurrence, likely due to reduced exposure to factors like overflowing river water. High-altitude regions characteristically boast porous soils, adept at soaking up water due to their open structure. Similarity, areas with higher NDVI values unveil denser vegetation, which helps capture water and mitigates soil erosion, ultimately reducing the likelihood of floods. It is worth noting that slope, TWI, and land use are not significantly correlated with flood occurrence. This may be attributed to the flat topography of Tianjin and the land use types dominated by agriculture and impervious roads.

Feature engineering serves to decrease the complexity of the model, allowing it to better concentrate on essential information, thereby enhancing the model's accuracy and generalization capability. Thus, we employ the FFIR method to further determine the optimal input features for the ML models based on the results of the correlation analysis. The ranking of features importance as shown in Fig. 7. According to the figure, the polarizations band of log|VH|, VV / VH, log|VV|, VV * VH, along with the feature of distance to river explain 90 % of the total feature importance. To minimize the interference of redundant features on the model, we used the median of feature importance scores over 100 iterations as a threshold. Features with importance scores exceeding this median were selected as the final set of model input variables, the result as shown in Table 2. The final determined input features for the flood model include log|*VH*|, VV / VH, log|*VV*|, VV * VH, distance to river, VV, VV+VH, VV^2 * VH^2, VV^2 + VH^2, and VH.

4.2. Evaluation metric performance of model

To obtain a reliable model for urban flooding area extraction with good performance and universality, four standalone ML models, including XGBoost, SVC, MLP and MDL were selected to compared with the EMF model. The statistical indicators of models on the training

Table 2
The optimal features for model development.

Sequence number	Feature name	Sequence number	Feature name
1	log|*VH*|	6	VV
2	VV / VH	7	VV+VH
3	log|*VV*|	8	VV^2 * VH^2
4	VV * VH	9	VV^2 + VH^2
5	*Distance to river*	10	VH

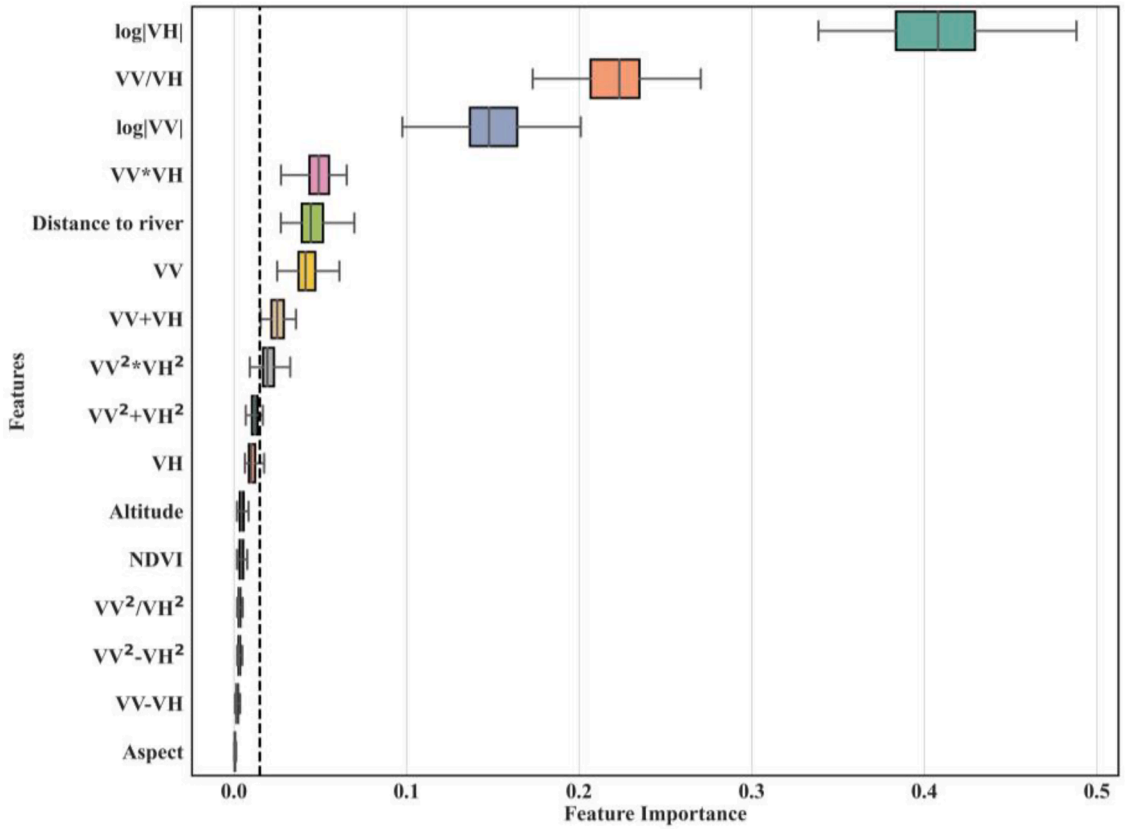

Fig. 7. The ranked importance of factors significantly associated with flooding. The outcomes from 100 (n) model iterations are depicted using box plots, illustrating the median values (vertical line within each box) and the interquartile range (width of the box).

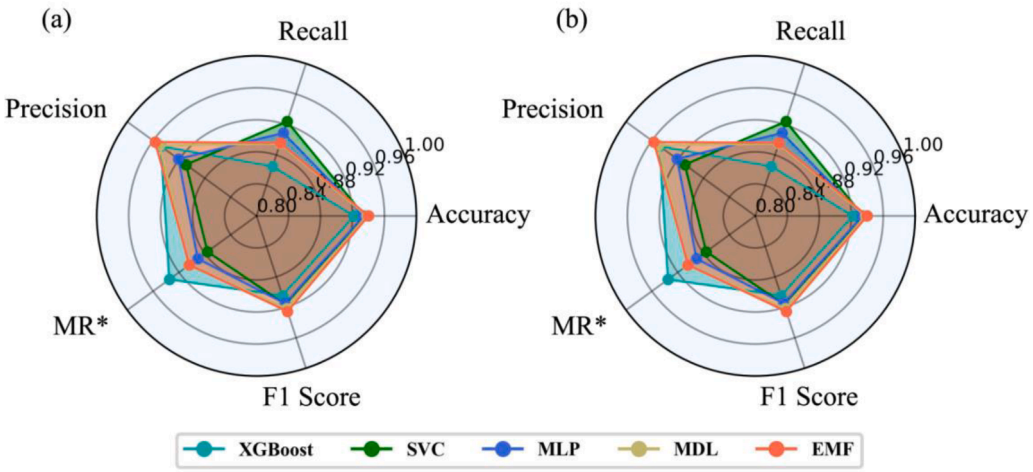

Fig. 8. Performance evaluation for the training dataset (a) and testing dataset (b) using the XGBoost, SVC, MLP, MDL and EMF model. To facilitate a visually comparative assessment of classification accuracy across different models on a uniform scale, the miss rate values of all models were adjusted by adding 0.8. The adjusted metric is denoted as MR* in the figure.

dataset and the testing dataset as show in Fig. 8. In general, all the ML algorithms obtained good accuracy in both the training dataset and testing dataset (accuracy> 0.9). Specially, the accuracy, precision and F1 score of the ML models in training and testing datasets ranks from high to low as: EMF > MDL > XGBoost > SVC > MLP. The EMF model exhibit the highest accuracy, precision and F1 score values among five models, indicating that it can effectively identify flooding regions. The values of accuracy, precision and F1 score for the EMF model in both the training and testing datasets, which are 0.94, 0.96 and 0.93, respectively, demonstrate the strong generalization of the EMF model. Compared to the standalone models, the EMF model exhibits moderate recall and MR* values, indicating that these two models are more conservative in identifying flood-prone areas with the goal of minimizing the risk of misclassification.

4.3. *Mapping performance of using different ML models*

Previous studies indicated that the accuracy of different classification models can have a significant impact on the accuracy of flood extraction spatial mapping (Luoto et al., 2010). Land cover types exhibit significant variations in spatial heterogeneity. For example, urban areas are highly fragmented, encompassing diverse land cover types within small areas. This heterogeneity presents challenges for flood models in accurately identifying and delineating flooded areas amidst mixed pixels. To investigate the EMF model's performance in extracting floods across diverse land cover types, we mapped the spatial distributions of flooded areas in mountainous regions, farmlands, and urban roads using the trained five machine learning models on July 31, 2023. This visual assessment allowed us to evaluate the EMF model's effectiveness in extracting the spatial extent of flooding in Tianjin.

According to Fig. 9 (a)–(e), the flood areas extracted by the five models exhibit high spatial consistency across farmlands. Farmlands typically have a relatively homogeneous land cover, consisting mainly of vegetation and bare soil. This homogeneity reduces the complexity of the SAR signal and makes it easier for models to distinguish flooded areas from other land cover types. Moreover, farmlands are typically open areas with minimal shadowing from trees or buildings. This lack of shadowing ensures a consistent signal across the farmland, further aiding in accurate flood detection. Nevertheless, standalone ML models still tend to underestimate the extent of flood areas in small villages within

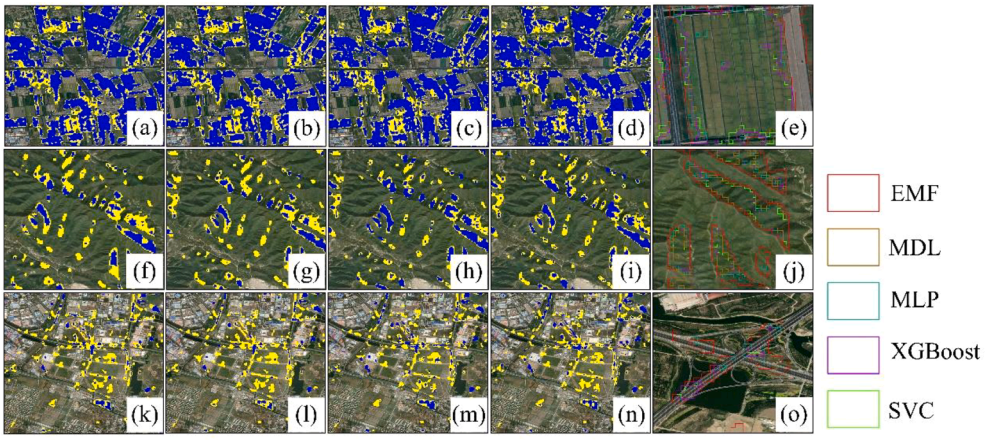

Fig. 9. Comparison of flood area extraction by four standalone ML models and the EMF model in farmland [(a)–(e)], mountain [(f)–(j)], and urban areas [(k)–(o)]. Blue color represents the extracted flood area by the standalone model. Yellow indicates the additional flood-affected areas identified by the EMF model. (a), (f), and (k) compare the flood areas extracted by XGBoost and EMF models. Similarly, (b), (g), and (l) compare SVC and EMF models; (c), (h), and (m) compare MLP and EMF models; and (d), (i), and (n) compare MDL and EMF models. (e), (j), and (o) depict the flood boundaries extracted by different models in locally zoomed-in areas of farmland, mountainous, and urban areas, respectively. The colors red, brown, blue, purple, and green represent the flood extents extracted by the EMF, MDL, MLP, XGBoost, and SVC, respectively.

agricultural fields and road areas between farmlands (Fig. 9 (a)–(d)). These areas typically feature abundant land cover information and intricate characteristics, which may challenge standalone ML models in accurately distinguishing floods from other surface features. In mountainous areas, all four standalone models underestimate the extent of flooding in valleys (Fig. 9 (f)–(j)). This is likely due to the fact that shadows cast on valleys by surrounding mountains can significantly affect the radar signal of SAR images. Standalone models may not be able to account for these shadowing effects properly, leading to underestimation of flooded areas. Additionally, the variable topography of mountainous areas, including steep slopes, canyons, and river valleys, leads to a complex and heterogeneous spatial distribution of floodwater. This complexity poses challenges for single models to accurately identify flood-affected areas. As shown in Fig. 9 (k)–(o), standalone models underestimate the extent of flooding in urban areas to a greater extent than in agricultural and mountainous areas. This is due to the similar backscattering characteristics of water and smooth road surfaces, where most of the incident radiation is reflected in the specular direction away from the radar. Additionally, urban structures such as buildings and bridges can cause multiple reflections and shadowing effects on radar signals, which may obscure the radar signatures of flood areas. Therefore, distinguishing between flooded and non-flooded areas within urban road networks can be challenging for standalone models. In the EMF model, the RFC model was employed to mitigate the discrepancies between predictions made by the standalone model and the data obtained through social sensing. By combining the predictions of several individual models, the EMF model exhibits improved noise resistance and reduced susceptibility to overfitting and errors that can occur in standalone models. This combination leads to more robust and reliable flood extraction performance, especially in complex environments like mountains and urban areas. Overall, the results demonstrated that the EMF model mapped the spatial distribution of the flood region with greater continuity and precision compared to other four standalone models.

4.4. Flood areas mapping

To further investigate the impact of model accuracy on flood area mapping, we compared the disparities in estimated total flood area by each trained ML model (Fig. 10). According to the figure, the EMF model estimated the largest flooded area at 1265.14 km^2, followed by the MDL (1029.29 km^2), MLP (981.46 km^2), and XGBoost (928.56 km^2). The flood area estimated by SVR was smallest among all the models, with the area of 907.59 km^2. The discrepancies in estimated flood areas highlight the variability inherent in ML model predictions, with standalone models tend to underestimate the extent of flood-affected areas.

The spatial distribution of flood area in Tianjin on July 31, 2023 was mapped by using the trained EMF model. As depicted in Fig. 10, the flood area is primarily located in the central-eastern and southern regions of the city. These regions situated in proximity to the Bohai Sea, the influx of humid monsoonal air masses from the eastern maritime direction, carrying a higher content of atmospheric moisture, leads to an increased occurrence of precipitation (Ren et al., 2022). In the western region of the city, the flood-affected area was smaller and relatively dispersed. Tianjin is bordered by mountains in the northern part of the city, rendering it a topographic trap for rainfall during the summer typhoon season. In the northern area, the downward flow of water results in flooding being primarily concentrated in the ridges and valleys.

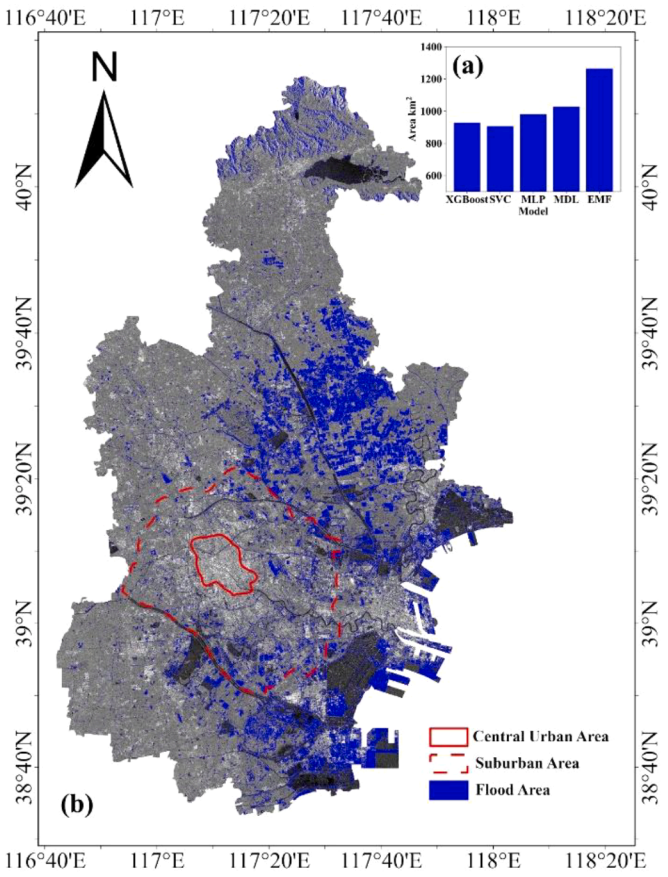

Fig. 10. (a) The comparison of total flood areas extracted by XGBoost, SVC, MLP, MDL and EMF models in Tianjin on July 31, 2023. (b) The spatial distribution of flood area in Tianjin using EMF model. The solid red line represents the central urban area, while the dashed line represents the suburban area. The gray base map corresponds to Sentinel-1 SAR data using the VV polarization band

5. Discussion

5.1. Sensitivity analysis

Sensitivity analysis is a widely adopted method for comprehending the impact of input variations on model outputs (Razavi et al., 2021). Thus, we applied sensitivity analysis to further assess the robustness capability of the EMF model. Specifically, we implemented four distinct training-testing configurations, utilizing 80 %, 70 %, 60 %, and 50 % of the total sample set for model training, with the remaining samples employed for model testing.

As shown in Fig. 11, the EMF model trained with a smaller sample exhibit the best stability among all models. More specially, in the training set, as the percentage of training datasets decreased from 70 % to 50 %, the accuracy of the XGBoost, SVC, MLP, MDN, and EMF model decreased by 0.013, 0.029, 0.030, 0.031and 0.002, respectively. In the testing set, the accuracy decreased by 0.014, 0.030, 0.035, 0.028 and 0.010, respectively. The data-driven models require a sufficient volume of sample information to effectively learn the non-linear relationships between input and output variables. Therefore, as the proportion of the training dataset decreases, there is a noticeable decline in the accuracy of the standalone models. The EMF model mitigates the risk of overfitting and enhances robustness and stability by harnessing the predictive capabilities of multiple models. Thus, compared with standalone models, the EMF model exhibited superior stability and better model performance.

5.2. Social sensing

The primary focus of a flood identification model is the precise identification of inundated land features. Since the accuracy and effectiveness of ML models are greatly dependent on the quantity and quality of data (L'Heureux et al., 2017), thus acquiring the flooding samples data is necessary. However, due to the adverse weather conditions and

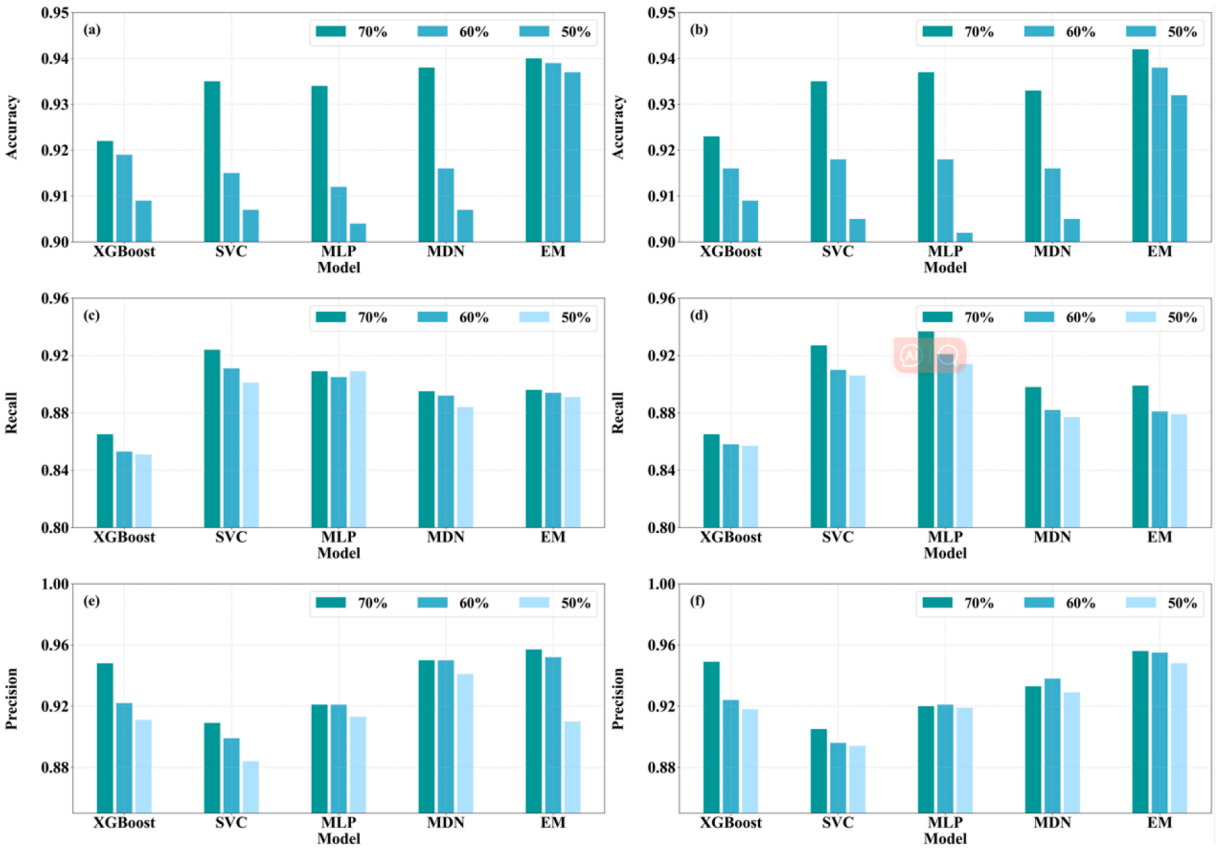

Fig. 11. The comparison of accuracy, recall, and precision for models in the training dataset (a, c, e) and the testing dataset (b, d, f), conducted using varying percentages of training data. Three simulation scenarios were set to train the XGBoost, SVC, MLP, MDL, and EMF models, respectively, using 70 %, 60 %, and 50 % of the total samples.

potential safety hazards during the extreme rainfall event, obtaining sample data for flood-affected areas using traditional on-site survey presents a considerable challenge. Social sensing encompasses a range of sensing and data collection approaches where data are gathered either by humans or mobile devices (Wang et al., 2015). The social sensing data has been utilized to enhance flood mapping by providing discrete inundation submerging information across study areas (Karmegam et al., 2021). Citizens frequently share local disaster information on platforms, which offers real-time, potentially accurate insights into local flood event.

Thus, in this study, the social sensing framework was utilized to identify information about flood inundation locations and enhance the model's capacity for learning flood submersion data. It worth to note that the social sensing data for flood monitoring can be noisy and inaccurate since the data is collected from a variety of sources, including social media, news articles, and government websites. This can lead to inconsistencies in the data format, quality, and accuracy. To ensure the accuracy of social sensing data for flood monitoring, this study employs rigorous data source filtering and data quality filtering standards. Firstly, the timing of social media posts should coincide with the occurrence of the flood. Secondly, video posts should visually depict instances of flooding, whereas text posts should be distributed through official media channels, such as the Tianjin Traffic Police. Data identified as originating from unreliable sources, such as bots or trolls, was deemed unreliable or inaccurate and was therefore excluded. Additionally, data missing key information, such as the date and location, is removed. Some of the social sensing results as depicted in Table 3. For example, the text description of a TikTok video posted on August 1, 2023, indicates that the accumulated water at the AEON Supermarket in the Xiqing District of Tianjin has overflowed onto the pedestrian walkway. Furthermore, the video clearly shows that the floodwaters have reached halfway up the bodies of several cars [Fig. A.2 (a)]. Based on the above information, we consider this to be valid social media data. Through the integration of the GEE API with geographical coordinates collected via social sensing, a total of 1500 flood sample data points were acquired.

Furthermore, we configured scenarios without including flood sample points to develop all five models, further exploring the necessity of social sensing. The results are shown in Fig. 12. In general, all the ML model exhibit a high level of accuracy (Accuracy $>$ 0.9) in both the training and testing datasets, indicating the effectiveness of the developed models in recognizing water bodies and land areas. However, the accuracy of all the ML models drops dramatically when identifying the flooding points. The accuracy score of XGBoost, SVC, MLP and MDL were 0.257, 0.298, 0.323, and 0.331 respectively. Even with the integration of ML models by using the RF model, the accuracy of the EMF model remains at 0.523. This indicates that the models without the social sensing dataset struggled to accurately recognize the submerged area of flooding areas. After incorporating data obtained through social sensing, each model shows improved performance in identifying flood-prone areas, demonstrating the effectiveness of social sensing.

5.3. *Flood risk disaster management*

Utilizing ERA5 data, we determined the average daily precipitation in Tianjin from January 2014 to November 2023. Applying the official standards of 25 mm and 50 mm to define heavy and rainstorm events, respectively, we identified a total of 51 days with heavy precipitation, including 9 rainstorm occurrences during this period (Fig. 13 (k)). Notably, these events primarily concentrated in July and August of each

Table 3
Example of social textual information related to flooding.

Post time	Address	Textual information	Video information	Platforms (post type)
2023.08.01	天津市大悦城 (Joy City, Tianjin)	涨水 (flooding)	汽车被洪水浸泡 (car is submerged in water)	TikTok (video)
2023.08.01	天津市西青区永旺超市 (The AEON Supermarket in Xiqing District, Tianjin)	积水漫过步行道 (Accumulated water over the sidewalk)	汽车被洪水浸泡 (cars are submerged in water)	TikTok (video)
2023.07.31	天津市滨海新区京山铁路桥 (Jingshan Railway Bridge, Binhai New Area, Tianjin)	因积水采取封闭措施 (Closure measures taken due to flooding)	/	Toutiao (text)
2023.07.31	天津市河西区爱国道 (Aiguo Road, Hexi District, Tianjin)	积水严重 (Severe water accumulation)	目视可见洪水 (floodwater is visible)	Kuaishou (video)
2023.08.02	天津市南开区南泥湾路 (Nannibei Road, Nankai District, Tianjin)	暴雨之后 (After the heavy rain)	汽车被洪水浸泡 (car is submerged in water)	TikTok (video)
2023.07.31	天津市大邱庄长江道 (Changjiang Road, Daqiu Zhuang, Tianjin)	积水严重 (Severe water accumulation)	目视可见洪水 (floodwater is visible)	Kuaishou (text)

year. Therefore, accurate and precise flood area extraction and delineation are crucial for effective disaster management and resource allocation, especially in light of the potential for significant flood events during these months. It enables authorities to plan and execute timely response measures, minimize property damage, and ensure the safety of affected populations. In this study, the Sentinel-1 SAR data was utilized to extract the extent of urban flooding. SAR data offer several advantages over traditional methods of flood monitoring, including all-weather capability and high spatial resolution. These advantages make SAR satellites valuable tools for flood monitoring, providing valuable information for emergency response and disaster management. We selected five Sentinel-1 SAR data sets associated with heavy rain events for spatial mapping. When comparing all the ML-derived flood area maps with land use data, we discovered that most of the flood-prone areas are located within artificial zones, particularly in agricultural areas (Fig. 13 (a) and (b)). Among the five flooded areas depicted in the maps, the percentage of inundated land that was agricultural land ranged from 54.2 % to 56.1 %, with an average of 52.8 %. The percentage of impervious surfaces in the total flooded area was 32.3 %, 32.3 %, 34.5 %, 33.0 %, and 32.8 %, respectively (Fig. 13 (f)–(j)). This highlights the flood risk to human life and the economy. In January 2023, the Tianjin Government issued a *Comprehensive Rural Revitalization Action Plan for Tianjin*, with a focus on prioritizing the development of livable, business-friendly, and aesthetically pleasing rural areas. Ensuring the effective safeguarding of agricultural resources is crucial for the smooth implementation of the policy. Chapman et al. (2022) found that effective green infrastructure and sustainable drainage systems can achieve multiple ecosystem benefits, providing valuable guidance for suburban flood management. Therefore, to mitigate the impact of floods on farmland, the government may consider enhancing infrastructure development in agricultural areas, such as levees and flood warning systems.

Furthermore, we have observed that flood disasters in outskirts areas tend to be more severe than in center urban areas and suburb areas. The percentage of outskirts areas in the total flooded area was 88.49 %, 87.97 %, 87.50 %, 87.03 %, and 79.53 %, respectively. Floods in suburb areas accounted for no more than 21 % of the total flooded area, and floods in center urban areas accounted for no more than 1 % of the total flooded area. This is due to the better drainage systems and flood mitigation measures in cities, such as flood walls, and drainage facilities. Thus, it is imperative for the government to develop and implement robust disaster preparedness plans and strategies specifically tailored to outskirt areas to effectively respond to and manage flood events. In addition, some studies have indicated that adopting Low Impact Development (LID) measures to replace impervious surface development could effectively address the challenges posed by climate change and alleviate urban flooding (Hassani et al., 2023; Mousavi Janbehsarayi et al., 2023). The government can integrate the sponge city concept into the future development of suburb and outskirts areas prone to flooding, implementing measures such as permeable pavements, green roofs, and stormwater disposal facilities to mitigate the occurrence of flood disasters in these areas (He et al., 2019; Wang et al., 2023).

Previous studies have underscored the increasing impact of extreme rainfall on the safe operation of urban transportation facilities, posing a significant challenge to public safety (Ding et al., 2023; He et al., 2023). Analyzing the flooding maps of Tianjin during extreme rainfall events, we identified low-lying bridges as one of the primary flooding locations within the center urban and suburb areas. Unlike open land, bridges have limited space for water to drain away. Although the drainage pumps have been installed at the downhill locations of the bridge, there is still have the high flooding hazard risk when strong and short-term rainfall occur. As a result, there will be threats to people's lives and

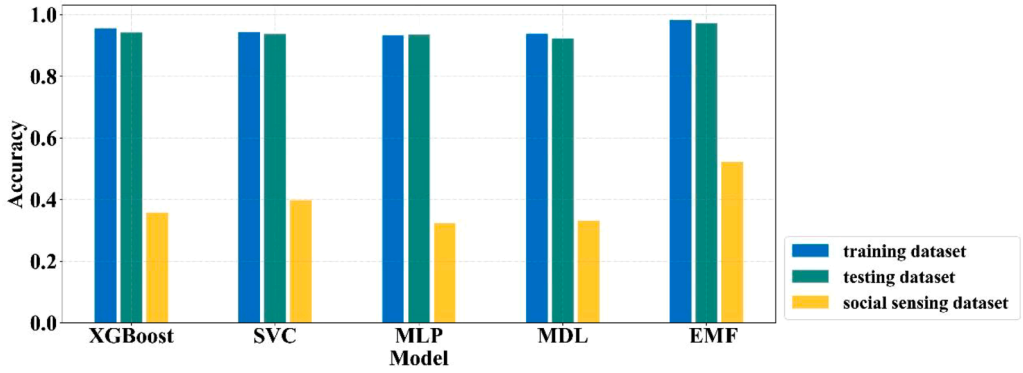

Fig. 12. The accuracy comparison of four standalone ML models and EMF model under the scenarios without social sensing data in training dataset and testing dataset, and the model's performance for predicting the social sensing dataset.

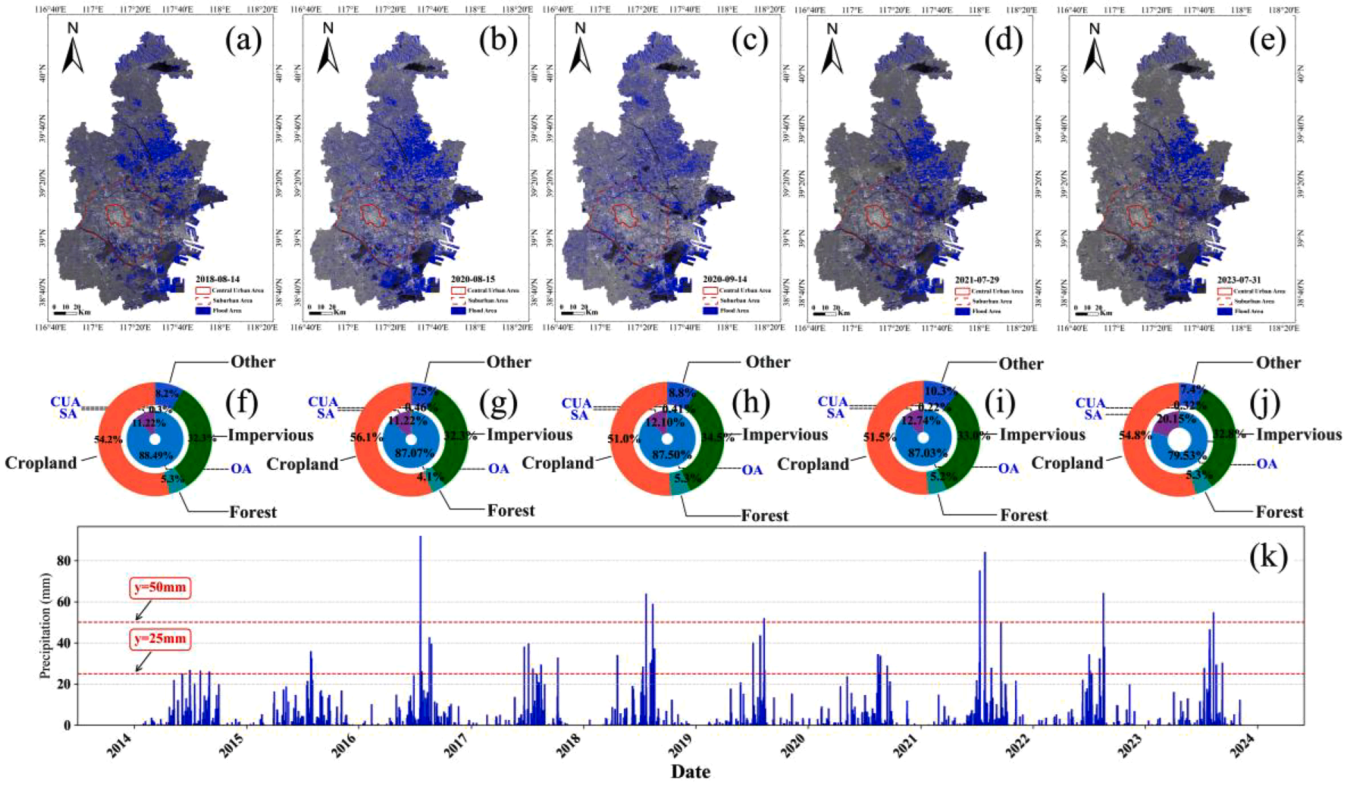

Fig. 13. Flood mapping for five heavy rain events from 2018 to 2023 using Sentinel-1 SAR data and the EMF model. (a) – (e) the spatial mapping during the on specific dates. (f) to (j) the proportional areas of flood coverage for different land-use types and center urban area (CUA), suburb area (SA), and outskirts area (OA) on the corresponding dates. (k) the average precipitation from January 2024 to November 31, 2023, acquired from ERA5 data. The dashed lines represent the officially defined criteria for heavy rain and rainstorm events.

property. Additionally, Rajput et al. (2023) found that the impact of floods on urban transportation networks is persistent, as networkwide increased travel times could last for weeks even after the floodwater has receded. Thus, strengthening the supervision of these locations during an extreme weather condition is crucial to the sustainable development of cities. Simultaneously, government agencies need to sustain their attention on the restoration of the functionality of transportation networks following flood disasters.

5.4. Comparison with previous studies

This study develops a technical framework for urban flood susceptibility mapping. To assess the efficacy of the proposed framework, Tianjin City, a coastal metropolis with a history of urban flooding, was selected as the typical study area. Apart from this study, some scholars have also conducted research on floods in Tianjin and have achieved significant results. For example, Li et al. (2016) proposed a multi-level and modular for simulating the urban flooding and tested it in a watershed of Hexi District, Tianjin. The results showed a relative error control of less than 5 %. Although the model exhibited good performance, being grounded in physics, it necessitates accurate external datasets, such as road networks and urban drainage systems. Furthermore, the precision of the model in simulating flood scenarios is notably impacted by urban topography and morphology, while its generalizability is constrained. Yan et al. (2021) developed a rapid urban flood inundation prediction model based on the PCSWMM and neural network models in a drainage area in Tianjin, with a Nash efficiency coefficient of 0.876. However, the model's portability was hindered by external data prerequisites, including the need for rainfall data and urban underlying surface feature data. Furthermore, the limited size of the dataset used (35 data samples) and the reliance on a single machine learning model constrained the enhancement of the model's accuracy. In our proposed framework, Sentinel-1 SAR data are accessed freely on the GEE platform. Furthermore, as a data-driven model, the EMF model enables the identification of intricate patterns and interrelationships within the data, particularly suitable for complex urban land cover scenarios. By combining multiple base classifiers, EMF model can mitigate the bias of individual classifiers. The results indicate that the EMF model exhibits high accuracy in identifying urban flood susceptibility areas, achieving accuracies of 0.94 and 0.96 in the training and testing datasets, respectively.

Currently, the scarcity of flooding information poses a challenge to flood research in Tianjin. Wang et al. (2022) employed a combined approach of multivariate stepwise regression and Boosted Regression Trees model to investigate the urban flood spatial characteristics of nine Chinese cities, including Tianjin, Shanghai, and Guangzhou. Due to the dataset used in the study not containing timestamps, it is unable to provide effective guidance for the real-time extraction of flood inundation areas. Li et al. (2024) utilized the MaxEnt software to simulate waterlogging points in the central area of Tianjin, China. The results indicate that the MaxEnt software can effectively identify high flood risk areas. However, due to the lack of observed flood data, the authors utilized the information on susceptible waterlogging points from urban planning report, which may have issues with data incompleteness and insufficient accuracy. Social media data possesses timeliness and comprehensiveness, enabling rapid acquisition of extensive user reports and feedback on flood events. Yan et al. (2024) proved the potential of utilizing social sensing textual data to capture the spatiotemporal dynamic changes of urban flooding. In this study, we leveraged video and text media data to further investigate the potential of social sensing in flood monitoring applications. The results indicated that the social media data can serve as a crucial supplement to remote sensing

observations, offering more detailed and precise flood information. Overall, compared to previous studies, the new framework proposed in this paper facilitates the rapid acquisition of reliable urban flood information and extensive monitoring of flood inundation areas, providing technical support for urban flood emergency management.

5.5. Model limitations

The EMF models introduced in this study also had some limitations. First, as a data-driven model, the performance of the EMF algorithm heavily relies on the quality of the input data. All ground truth data used for the model training and validation were obtained from the social sensing method and visual interpretation method. Although these data sets covered sufficient flooding samples and no-flooding samples of Tianjin in the extreme rainfall event, the lack of training and validation data of other cities would lead to weak model generalization. Therefore, the EMF model proposed in this study were considered effective on a regional scale, and it is necessary to employ new data for additional model calibration before extending their application to different regions. Second, the ensemble model adeptly capitalizes on the strengths of individual models, thereby diminishing the variance of a single model and alleviating the risk of overfitting. Consequently, this augmentation leads to an enhancement in the predictive performance of the model. However, the challenge of determining which ML models to employ, how to configure them optimally, and how to orchestrate their collective decision-making poses a considerable hurdle in the development of a robust flood recognition model. Among the multitude of ML and DL models, XGBoost renowned for its high predictive accuracy and feature selection capabilities (Cherif & Kortebi, 2019). SVC excels in nonlinear data classification and is robust in high-dimensional spaces (Singla et al., 2020). MLP adept at complex tasks with adaptive feature learning (Schmidhuber, 2015). MDL effectively integrates multiple data modalities (John et al., 2023). Therefore, we select these four models as the initial components of the EMF model, providing independent prediction results. Subsequently, we employ RFC to couple the results of these four models. The choice of RFC is due to its high tolerance for diverse models and the advantage of requiring fewer hyperparameters (Belgiu & Drăguţ, 2016). Nevertheless, due to the "black box" nature of machine learning models, it is imperative to conduct a thorough assessment of the strengths and weaknesses of individual models when applying the EMF model to other regions. In cases where one of the four standalone models proves ineffective, the consideration of a suitable replacement model should be contemplated. Thirdly, compared with standalone ML models, the EMF model have more complex structures and hyperparameters. Besides, it requires a large dataset for model training. The high requirement computational expenses and resources also pose a challenge to the model development process.

6. Conclusions

It is significant for ensuring the safety of residents' lives and property to monitor flooding and assess the flood disasters risk during an extreme precipitation event. In this study, a hybrid ML model, named the EMF model, was developed to identify flood-prone areas using Sentinel-1 SAR imagery. By using the feature selection and multiple model ensemble strategy, the EMF model can further improve classification accuracy compared to standalone ML models. By introducing sensitivity analysis, our results showed that the EMF model exhibits robust performance across training datasets of varying proportions. Furthermore, the EMF model demonstrates practical applicability in satellite-based applications, showing higher accuracy in extracting flood inundation extents across various land cover types compared to other models. The flood-inundated areas of Tianjin on July 31, 2023 were ultimately mapped using the EMF model and Sentinel-1 SAR imagery. The results showed that the majority of flood-inundated areas in Tianjin are concentrated in suburban and mountainous regions. It indicates that the flood prevention and drainage measures in the city of Tianjin are well-implemented. However, flood prevention measures are still needed for both urban overpass areas and rural areas. Overall, the scheme proposed in this paper shows the potential of machine learning models and satellite data to identify the flood-prone areas, which could help decision-making departments in implementing effective flood response measures and initiating material rescue operations.

In terms of future work, this study will primarily focus on three aspects. First, integrating additional data sources with remote sensing data and social sensing data, such as historical flood records, water level, and road sensor data, to collect more reliable flooding sample data. Second, incorporating more ML and deep learning models into the ensemble ML framework to further enhance the accuracy and performance of urban flood susceptibility mapping. Third, evaluate the transferability and scalability of the proposed methodology to other urban areas to support the adoption of flood mapping techniques across diverse geographical contexts.

CRediT authorship contribution statement

Xiaotong Zhu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hongwei Guo:** Writing – review & editing, Validation, Methodology, Data curation. **Jinhui Jeanne Huang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgments

The authors acknowledge the financial support of the National Key R&D Program of China for this research under grant number 2021YFC3200404.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.scs.2024.105508.

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